

Name: Omer Boztepe, Mahir Camci

CMPM 118

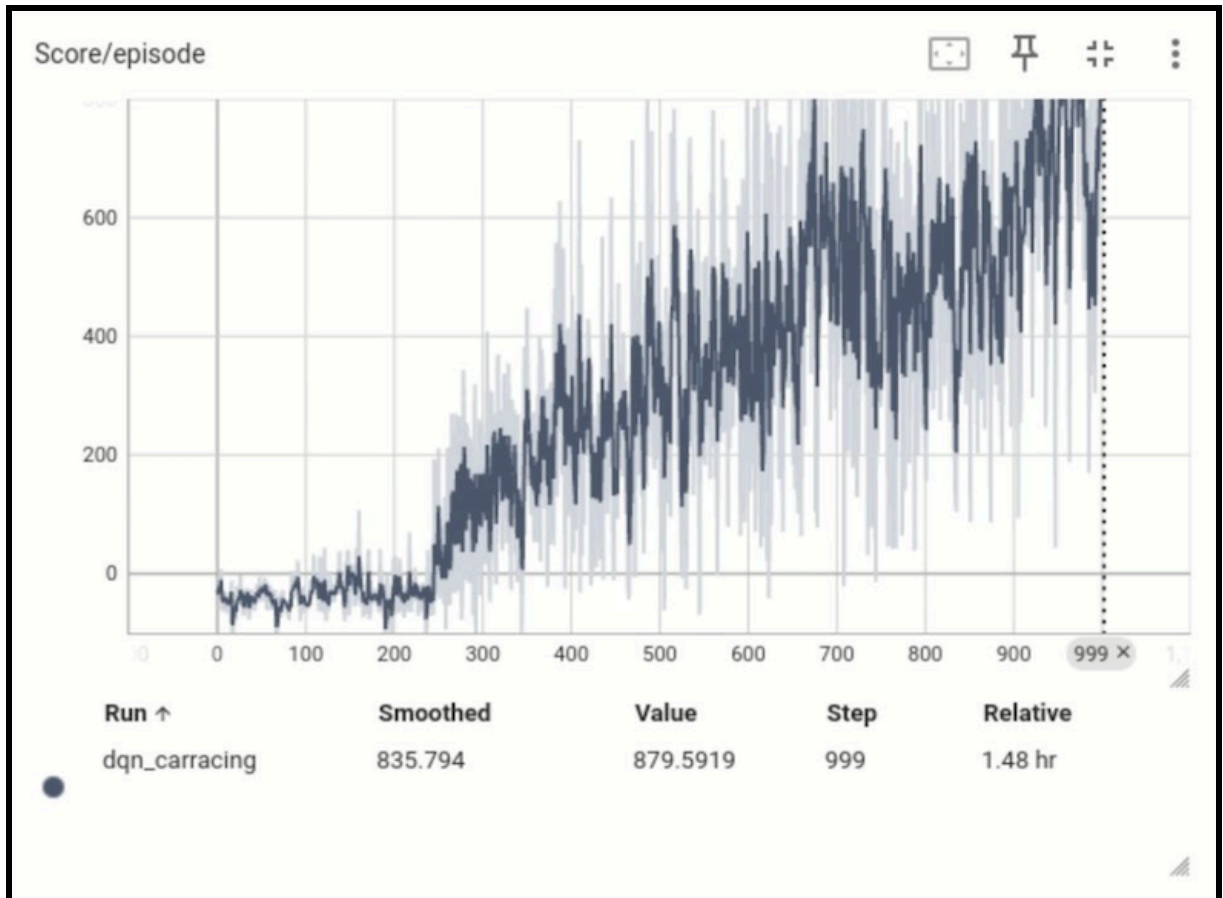
April 19, 2026

Task 8

After finishing Task 7 where we made a DQN algorithm that gave us satisfactory results, we moved to Task 8 to further improve our algorithm. We checked a few websites and YouTube videos to find ways to develop. We also received help from LLMs to maximize our chances of properly completing the task. Here is one of the places that we got inspiration from:

<https://github.com/wiitt/DQN-Car-Racing>.

To try and achieve potential improvements, several parts of our DQN implementation were changed. The major architectural change was to move from the 2 layered convolutional network to a 3 layered network ($32 > 64 > 64$ filters) with a bigger fully connected layer (512 units instead of 256), making it similar to the standard Atari DQN architecture. The input size was decreased to 48×48 from 84×84 , to decrease the time taken for training, and the action space was reduced from 7 to 5 by removing diagonal steer + gas actions. The optimizer was also changed from RMSprop to Adam with MSE loss. Epsilon was annealed multiplicatively on an exponential decay, instead of linearly from task 7. The reward shaping used for task 7 was removed, which was not surprising given that our original method had never been tested with the original reward function (except with the standard DQN which had shown a dramatic performance improvement when reward shaping was used). Here are the results we have:



dqn : bash — Konsole

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```
episode: 970 score: 750.00 avg score: 644.15 epsilon: 0.05 steps: 250 fps: 45
episode: 971 score: 877.42 avg score: 646.67 epsilon: 0.05 steps: 250 fps: 45
episode: 972 score: 896.54 avg score: 648.90 epsilon: 0.05 steps: 250 fps: 46
episode: 973 score: 592.06 avg score: 651.01 epsilon: 0.05 steps: 250 fps: 45
episode: 974 score: 851.39 avg score: 658.66 epsilon: 0.05 steps: 250 fps: 45
episode: 975 score: 876.19 avg score: 660.12 epsilon: 0.05 steps: 250 fps: 44
episode: 976 score: 874.11 avg score: 664.80 epsilon: 0.05 steps: 250 fps: 45
episode: 977 score: 854.70 avg score: 667.74 epsilon: 0.05 steps: 250 fps: 47
episode: 978 score: 929.40 avg score: 671.17 epsilon: 0.05 steps: 177 fps: 46
episode: 979 score: 519.35 avg score: 671.97 epsilon: 0.05 steps: 250 fps: 45
episode: 980 score: 492.95 avg score: 673.73 epsilon: 0.05 steps: 250 fps: 45
episode: 981 score: 628.87 avg score: 676.53 epsilon: 0.05 steps: 250 fps: 47
episode: 982 score: 781.58 avg score: 678.50 epsilon: 0.05 steps: 250 fps: 45
episode: 983 score: 503.70 avg score: 680.19 epsilon: 0.05 steps: 250 fps: 46
episode: 984 score: 644.77 avg score: 683.75 epsilon: 0.05 steps: 250 fps: 47
episode: 985 score: 169.84 avg score: 676.44 epsilon: 0.05 steps: 250 fps: 47
episode: 986 score: 798.20 avg score: 681.71 epsilon: 0.05 steps: 250 fps: 44
episode: 987 score: 334.46 avg score: 683.05 epsilon: 0.05 steps: 250 fps: 50
episode: 988 score: 878.06 avg score: 683.06 epsilon: 0.05 steps: 250 fps: 45
episode: 989 score: 382.99 avg score: 678.45 epsilon: 0.05 steps: 250 fps: 45
episode: 990 score: 353.64 avg score: 676.54 epsilon: 0.05 steps: 250 fps: 44
episode: 991 score: 678.50 avg score: 679.31 epsilon: 0.05 steps: 250 fps: 45
episode: 992 score: 303.97 avg score: 673.68 epsilon: 0.05 steps: 250 fps: 45
episode: 993 score: 920.20 avg score: 677.37 epsilon: 0.05 steps: 200 fps: 46
episode: 994 score: 797.06 avg score: 679.98 epsilon: 0.05 steps: 250 fps: 48
episode: 995 score: 821.50 avg score: 680.70 epsilon: 0.05 steps: 250 fps: 46
episode: 996 score: 567.80 avg score: 680.83 epsilon: 0.05 steps: 250 fps: 46
episode: 997 score: 900.90 avg score: 687.36 epsilon: 0.05 steps: 248 fps: 47
episode: 998 score: 866.56 avg score: 687.77 epsilon: 0.05 steps: 250 fps: 46
episode: 999 score: 879.59 avg score: 689.40 epsilon: 0.05 steps: 250 fps: 45
(boztepetask5) ubuntu@boztepe-desktop:~/persistent/dqn$
```

Overall we were unable to get meaningful gains compared to task 7, the algorithm reached an average of around 689 by the 999 th episode, compared to 836 with the previous algorithm. As can be observed from the graph the agent starts learning from roughly episodes https://docs.google.com/document/u/0/?authuser=0&usp=docs_web200-300 and continues to rise, but then plateaus between episodes 600-700, with high variation between each episode for the remaining run time. Although some episodes reach a high score of 920s, this does not make a large enough improvement. We suspect that some of the changes may have harmed rather than helped the algorithm, not including the tanh reward shaping may have resulted in earlier training to be harder as the raw reward of CarRacing is more noisy, the very small replay buffer (10k vs 300k) means it cannot have access to a variety of training information and going back from double dqn to plain dqn may cause overestimation of rewards. This also significantly reduced the run time from Task 7 to 1 hour and 29 minutes, largely because of the reduction in image size.

Here is the link to the Github: https://github.com/OmerBoztepe/Task_8